

Online Exam Portal

Software Requirements Specification (SRS)

Software Engineering Mini-Project — Deliverables Part-1

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Revision History

Version	Date	Author	Description
1.0	28-Sep-2026	[Student Name(s)]	Initial release of the SRS for the Online Exam Portal.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional, non-functional and security requirements of the **Online Exam Portal (OEP)**, a web-based system for creating, conducting and evaluating examinations online. It is written for the development team, the testers, the course instructor and any stakeholder who needs to verify that the finished product satisfies the agreed needs. It is the baseline document from which the Test Plan and the Software Architecture & Design Specification are derived.

1.2 Scope

The OEP allows institutions to conduct objective (MCQ and True/False) examinations over the web. Teachers build a question bank and schedule exams; students attempt the exams within a timed window; the system evaluates answers automatically and publishes results; administrators manage users and monitor the system.

In scope: user registration and authentication, user management, question bank, exam creation and scheduling, timed exam attempt, automatic evaluation, results and reports, audit logging.

Out of scope (this release): subjective/essay answer grading, live video proctoring, payment processing, native mobile applications, integration with an external Learning Management System.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
OEP	Online Exam Portal — the system specified in this document.
SRS	Software Requirements Specification.
FR / NFR / SR	Functional Requirement / Non-Functional Requirement / Security Requirement.
MCQ	Multiple-Choice Question.
RBAC	Role-Based Access Control.
JWT	JSON Web Token used for stateless authentication.
TLS	Transport Layer Security.
RPO / RTO	Recovery Point Objective / Recovery Time Objective.
Attempt	One sitting of a student for one exam, from start to submission.
Question Bank	A repository of questions owned by teachers from which exams are composed.

1.4 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- ISO/IEC/IEEE 29148:2018, Systems and software engineering — Requirements engineering.
- IEEE Std 829-2008, Standard for Software and System Test Documentation.
- OWASP Top 10 (2021) and OWASP Application Security Verification Standard (ASVS) 4.0.
- Course document “Software Engineering Mini-Project Deliverables Part-1”, 2026.

1.5 Overview of the Document

Section 2 gives the overall description of the product. Section 3 lists the specific requirements: external interfaces, functional requirements, UML use cases, non-functional requirements and security requirements. Section 4 explains how each requirement will be verified and traced. Requirement identifiers (FR-xx, NFR-xx, SR-xx) are unique and are reused in the Test Plan and the Design Specification.

2. Overall Description

2.1 Product Perspective

The OEP is a new, self-contained, three-tier web application: a browser-based client, a REST application server and a relational database. It uses an external SMTP e-mail service only to send password-reset messages. It does not depend on any existing system.

2.2 Product Functions

- Account management: register, log in, reset password, manage users and roles.
- Question bank management: create, edit, delete and categorize questions.
- Exam management: create, schedule, publish and assign exams to student groups.
- Exam taking: timed attempt with navigation, review marking, autosave and submission.
- Automatic evaluation and result publication with optional answer review.
- Reporting: exam-wise result statistics, CSV export and administrator summary.
- Auditing: security-relevant events are recorded in an audit log.

2.3 User Classes and Characteristics

User class	Description	Technical skill	Frequency of use
Student	Registered candidate who attempts exams and views results.	Basic web usage	Several times per term
Teacher / Examiner	Creates questions and exams and reviews results.	Basic to moderate	Weekly
Administrator	Manages accounts, roles and monitors the system.	Moderate	Daily

2.4 Operating Environment

- Client: any device with a modern web browser (see NFR-06).
- Server: Linux server or container runtime with Java 17, Spring Boot 3, PostgreSQL 15 and Redis 7.
- Network: HTTPS over the Internet or campus intranet.

2.5 Design and Implementation Constraints

- The system shall be developed as a web application using open-source technologies.
- Personal data shall be handled in line with the institution's data-protection policy.
- The project must be delivered within the academic term by the student team.

2.6 Assumptions and Dependencies

- Users have a working internet connection and a valid e-mail address.
- An SMTP service is available to deliver password-reset e-mails.
- Student groups (classes) are created by the Administrator or Teacher before exams are assigned.
- Exams contain objective questions only, so automatic evaluation is always possible.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The client shall be a responsive single-page web application with role-specific dashboards. Screens: Login/Register, Student Dashboard, Exam Instructions, Exam Attempt, Result, Teacher Dashboard, Question Bank, Exam Editor, Reports, Admin User Management. Error messages shall be displayed in plain language next to the field concerned.

3.1.2 Hardware Interfaces

No special hardware interface is required. Users need a computer, tablet or smartphone with a keyboard or touch input.

3.1.3 Software Interfaces

Interface	Purpose	Protocol
PostgreSQL 15	Persistent storage of users, questions, exams, attempts and results.	JDBC over TLS
Redis 7	Storage of token blacklist and login-rate counters.	Redis protocol
SMTP server	Sending password-reset e-mails.	SMTP with STARTTLS

3.1.4 Communication Interfaces

The client and server communicate through a RESTful JSON API over HTTPS (TLS 1.2+). The API is versioned under the prefix `/api/v1`.

3.2 Functional Requirements

Each functional requirement is written as a single verifiable “shall” statement with a unique identifier. Priority: High = must have, Medium = should have, Low = nice to have.

ID	Requirement	Priority
FR-01	The system shall allow a new student to register by providing full name, a unique email address and a password that meets the password policy (minimum 8 characters, at least one upper-case letter, one digit and one special character).	High
FR-02	The system shall authenticate a user with email and password and shall redirect the user to the dashboard for his/her role (Student, Teacher, Administrator) within 3 seconds of successful login.	High
FR-03	The system shall temporarily lock a user account for 15 minutes after 5 consecutive failed login attempts.	High
FR-04	The system shall allow a user to reset a forgotten password through a single-use link sent to the registered email address; the link shall expire 30 minutes after it is issued.	Medium
FR-05	The system shall allow an Administrator to create, edit, deactivate and delete user accounts and to assign one role to each account.	High
FR-06	The system shall allow a Teacher to create an exam by specifying title, subject, duration (5–240 minutes), start date-time, end date-time, total marks and passing marks.	High
FR-07	The system shall allow a Teacher to add, edit and delete questions in a question bank; supported types are single-choice MCQ (2–6 options), multiple-select MCQ and True/False, each with a marks value and a correct answer.	High
FR-08	The system shall allow a Teacher to select questions from the question bank for an exam and to configure negative marking (0 to 100 % of the question marks).	High
FR-09	The system shall allow a Teacher to publish or unpublish an exam and to assign it to one or more student groups; only published exams shall be visible to assigned students.	High

ID	Requirement	Priority
FR-10	The system shall display to a Student the list of assigned exams grouped as Upcoming, Available and Completed.	High
FR-11	The system shall allow a Student to start an exam only between its start and end date-time and only if the number of attempts made is below the attempt limit (default: 1).	High
FR-12	The system shall display a countdown timer during an attempt and shall automatically submit the attempt when the allotted time expires.	High
FR-13	The system shall automatically save the answers of an in-progress attempt at least once every 30 seconds.	High
FR-14	The system shall allow a Student to move to any question, change an answer, and mark a question for review before submission.	Medium
FR-15	The system shall allow a Student to submit an attempt manually after a confirmation prompt; a submitted attempt shall not be editable.	High
FR-16	The system shall automatically evaluate all objective answers on submission and shall compute the total score applying the configured marks and negative marking.	High
FR-17	The system shall show the Student his/her score, percentage and Pass/Fail status within 5 seconds of submission, and per-question review only if the Teacher has enabled it.	High
FR-18	The system shall allow a Teacher to view result reports of his/her exams (average, highest, lowest, pass percentage) and to export them as a CSV file.	Medium
FR-19	The system shall allow an Administrator to view a summary report of the total number of users, exams and attempts.	Low
FR-20	The system shall randomize the order of questions and options per Student when the Teacher enables randomization for the exam.	Medium
FR-21	The system shall record an audit log entry (user, action, timestamp, IP address) for every login, logout, exam start, exam submission and user-management action.	High

3.3 Use Cases (UML)

3.3.1 Use Case Diagram

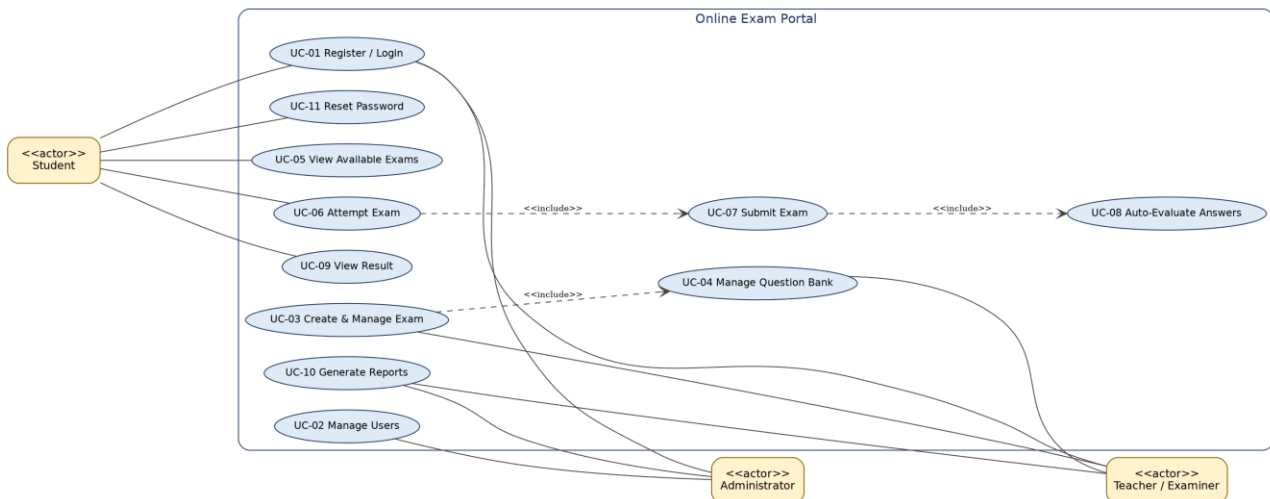


Figure 1: UML use case diagram of the Online Exam Portal

3.3.2 Use Case Summary

ID	Use case	Primary actor	Related FRs
UC-01	Register / Login	Student, Teacher, Administrator	FR-01, FR-02, FR-03, FR-21
UC-02	Manage Users	Administrator	FR-05, FR-21
UC-03	Create & Manage Exam	Teacher	FR-06, FR-08, FR-09, FR-20

ID	Use case	Primary actor	Related FRs
UC-04	Manage Question Bank	Teacher	FR-07
UC-05	View Available Exams	Student	FR-10
UC-06	Attempt Exam	Student	FR-11, FR-12, FR-13, FR-14
UC-07	Submit Exam	Student	FR-15, FR-21
UC-08	Auto-Evaluate Answers	System	FR-16
UC-09	View Result	Student	FR-17
UC-10	Generate Reports	Teacher, Administrator	FR-18, FR-19
UC-11	Reset Password	Student, Teacher, Administrator	FR-04

3.3.3 Use Case Specification — UC-01 Login

Actors	Student, Teacher, Administrator
Preconditions	The user has a registered and active account.
Trigger	The user opens the login page and submits credentials.
Main flow	1. User enters email and password. 2. System validates the input format. 3. System verifies the credentials. 4. System issues an authentication token. 5. System redirects the user to the role dashboard and writes an audit entry.
Alternative flow	3a. Credentials invalid: system shows “Invalid email or password”, increments the failed-attempt counter and returns to step 1. 3b. Fifth consecutive failure: account is locked for 15 minutes (FR-03).
Postconditions	User is authenticated with a token that expires after 30 minutes of inactivity (SR-04).
Requirements	FR-02, FR-03, FR-21, SR-01, SR-04, SR-06

3.3.4 Use Case Specification — UC-06 Attempt Exam

Actor	Student
Preconditions	Student is logged in; the exam is published, assigned to the student and the current time is within the exam window; attempts made < attempt limit.
Trigger	Student selects “Start Exam” on an available exam.
Main flow	1. System verifies eligibility. 2. System creates the attempt and starts the server-side timer. 3. System displays questions (randomized if enabled) without correct answers. 4. Student answers, navigates and marks questions for review; system autosaves every 30 s. 5. Student submits, or the timer expires and the system submits automatically (UC-07).
Alternative flow	1a. Not eligible: system shows the reason (not started, closed, attempt limit reached). 4a. Connection lost: on reconnection within 5 minutes the attempt resumes with saved answers (NFR-07).
Postconditions	Attempt is stored with status SUBMITTED and is passed to evaluation (UC-08).
Requirements	FR-11 to FR-15, NFR-07, SR-07, SR-09

3.4 Non-Functional Requirements

Every non-functional requirement contains a measurable acceptance criterion so that it can be tested objectively.

ID	Category	Requirement (measurable)	Priority
NFR-01	Performance	The system shall respond to at least 95 % of page and API requests within 2 seconds when 500 users are concurrently active.	High
NFR-02	Performance	The system shall compute and store the result of an exam with up to 100 questions within 5 seconds of submission.	High

ID	Category	Requirement (measurable)	Priority
NFR-03	Availability	The system shall be available at least 99.5 % of the time during scheduled exam windows, measured monthly.	High
NFR-04	Scalability	The system shall support 1,000 concurrent exam attempts by adding application instances, without a code change.	Medium
NFR-05	Usability	At least 90 % of first-time students shall be able to log in and start an exam within 3 minutes without training (tested with a minimum of 5 users).	Medium
NFR-06	Compatibility	The web client shall work on the latest two major versions of Chrome, Firefox, Edge and Safari and shall be usable on screens from 360 px width.	Medium
NFR-07	Reliability	After a network interruption of up to 5 minutes, a Student shall be able to resume the attempt with no loss of answers saved before the interruption.	High
NFR-08	Maintainability	The source code shall have automated unit-test coverage of at least 70 % and shall follow a modular layered structure.	Medium
NFR-09	Recoverability	The system shall take a full database backup every 24 hours ($RPO \leq 24$ h) and shall be restorable within 4 hours ($RTO \leq 4$ h).	Medium
NFR-10	Accessibility	All student-facing pages shall meet WCAG 2.1 Level AA (keyboard navigation, text contrast ratio $\geq 4.5:1$).	Low

3.5 Security Requirements

3.5.1 Security Objectives

ID	Objective	Description
SO-1	Confidentiality	Protect credentials, personal data, exam questions and answer keys from unauthorized disclosure.
SO-2	Integrity	Ensure that exam content, submitted answers and results cannot be altered by unauthorized parties.
SO-3	Availability	Keep the portal available and resistant to abuse (brute force, request flooding) during exam windows.
SO-4	Accountability	Make every security-relevant action traceable to a user through tamper-evident audit records.
SO-5	Fair Assessment	Prevent impersonation and unauthorized access to exams so that results reflect the candidate's own attempt.

3.5.2 Security Requirements

ID	Objective(s)	Requirement	Priority
SR-01	SO-1	User passwords shall be stored only as salted bcrypt hashes with cost factor ≥ 12 ; plaintext passwords shall never be logged or stored.	High
SR-02	SO-1	All client-server communication shall use HTTPS with TLS 1.2 or higher; HTTP requests shall be redirected to HTTPS.	High
SR-03	SO-1, SO-2	The server shall enforce role-based access control on every API call; a request for a resource not permitted to the caller's role shall be rejected with HTTP 403.	High
SR-04	SO-5	Authentication tokens (JWT) shall expire after 30 minutes of inactivity and shall be invalidated on logout.	High
SR-05	SO-2	All user input shall be validated on the server and database access shall use parameterized queries to prevent SQL injection and cross-site scripting.	High
SR-06	SO-3	The system shall limit login requests to 5 per minute per IP address and return HTTP 429 when the limit is exceeded.	High
SR-07	SO-1, SO-5	Correct answers shall never be sent to the client before the attempt is submitted.	High
SR-08	SO-4	Audit logs shall be append-only, retained for at least 1 year, and accessible only to Administrators.	Medium

ID	Objective(s)	Requirement	Priority
SR-09	SO-2, SO-5	The server-side clock shall be the authoritative exam timer; answers received after the attempt end time (plus 10 seconds grace) shall be rejected.	High
SR-10	SO-1, SO-2, SO-3	The release build shall be checked against the OWASP Top 10 (2021) with no open High or Critical findings.	Medium

3.5.3 Threats Considered

Threat	Addressed by
Credential theft / brute-force login	SR-01, SR-06, FR-03
Eavesdropping on network traffic	SR-02
Privilege escalation (student accessing teacher functions)	SR-03
Session hijacking / stolen token reuse	SR-04
SQL injection and cross-site scripting	SR-05, SR-10
Answer-key leakage and time manipulation	SR-07, SR-09
Repudiation of actions	SR-08, FR-21

4. Requirements Verification and Traceability

Each requirement is verified by at least one test case defined in the Test Plan (document OEP-TP-001, Section 13 “Traceability Matrix”). Verification methods: **Test** (execution against the running system), **Inspection** (review of code or configuration) and **Demonstration**.

Requirement group	Primary verification method	Test Plan reference
FR-01 to FR-21	Test (functional, manual and automated)	TC-01 to TC-09, TC-15, TC-17 to TC-21
NFR-01, NFR-02, NFR-04	Test (load test with JMeter)	TC-10
NFR-05, NFR-06, NFR-10	Test / Demonstration (usability session, browser matrix)	TC-14
NFR-07	Test (network interruption)	TC-15
NFR-03, NFR-08, NFR-09	Inspection (monitoring, coverage report, backup drill)	Section 5 of Test Plan
SR-01 to SR-10	Test and Inspection (security validation)	TC-03, TC-08, TC-11, TC-12, TC-13, TC-16, TC-19, TC-22, TC-23

Appendix A: Requirement Quality Checklist

Every requirement in this SRS was reviewed against the following criteria:

Criterion	How it is met
Clear and unambiguous	One “shall” statement per requirement; terms defined in Section 1.3.
Concise	Each requirement is limited to one behaviour or one quality attribute.
Measurable / testable	Numeric limits (time, count, percentage) are stated; each has a test in the Test Plan.
Consistent	Unique IDs; no two requirements contradict one another.
Traceable	IDs are reused in the Test Plan and the Design Specification.